

Jacobian Descent

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github.com/TorchJD/torchjd



torchjd.org



arxiv.org/pdf/2406.16232

Motivation

Partial order

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, as

$$\begin{aligned} \mathbf{u} \preceq \mathbf{v} \\ \Leftrightarrow u_i \leq v_i \quad \forall i \in [m] \end{aligned}$$

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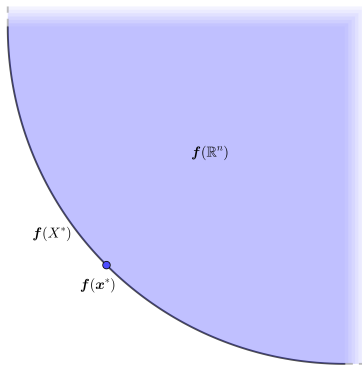
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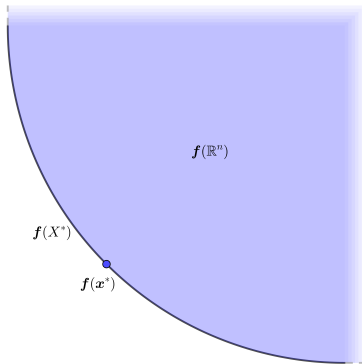
▶ $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

Pareto optimality



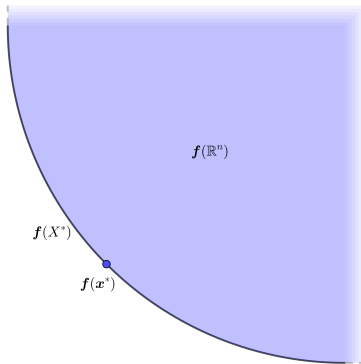
Pareto optimality

Pareto set: X^*



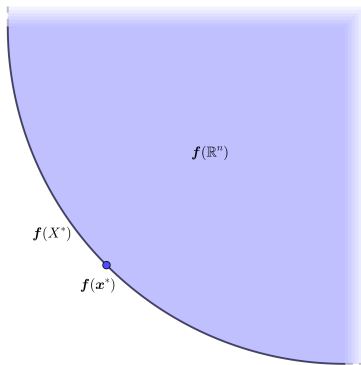
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$$\mathbf{x}^* \in X^*$$

$$\Leftrightarrow$$

$$\forall \mathbf{y} \in \mathbb{R}^n, \mathbf{f}(\mathbf{y}) \preceq \mathbf{f}(\mathbf{x}^*) \rightarrow \mathbf{f}(\mathbf{y}) = \mathbf{f}(\mathbf{x}^*)$$

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$$\mathcal{J}\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \nabla f_1(\mathbf{x})^\top \\ \nabla f_2(\mathbf{x})^\top \\ \vdots \\ \nabla f_m(\mathbf{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\mathbf{x}) & \frac{\partial}{\partial x_2} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\mathbf{x}) \\ \frac{\partial}{\partial x_1} f_2(\mathbf{x}) & \frac{\partial}{\partial x_2} f_2(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\mathbf{x}) & \frac{\partial}{\partial x_2} f_m(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\mathbf{x}) \end{bmatrix}$$

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We select $\mathbf{y} = -\eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x})) \in \mathbb{R}^n$.

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Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

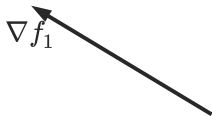
Input: $\mathbf{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$

for $t \leftarrow 1$ **to** T **do**

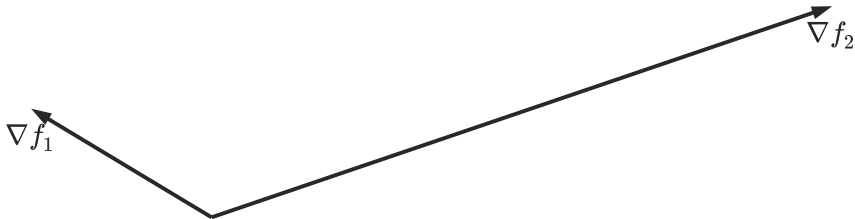
$\mathbf{x} \leftarrow \mathbf{x} - \eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x}))$

Output: $\mathbf{x}$

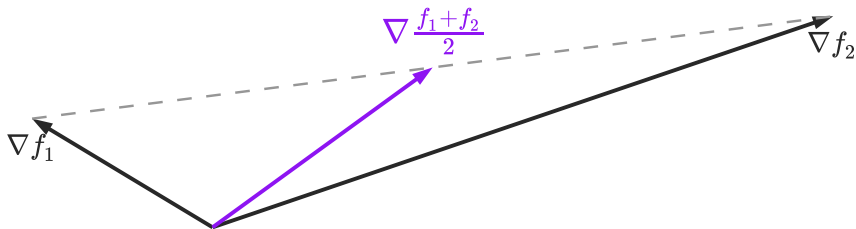
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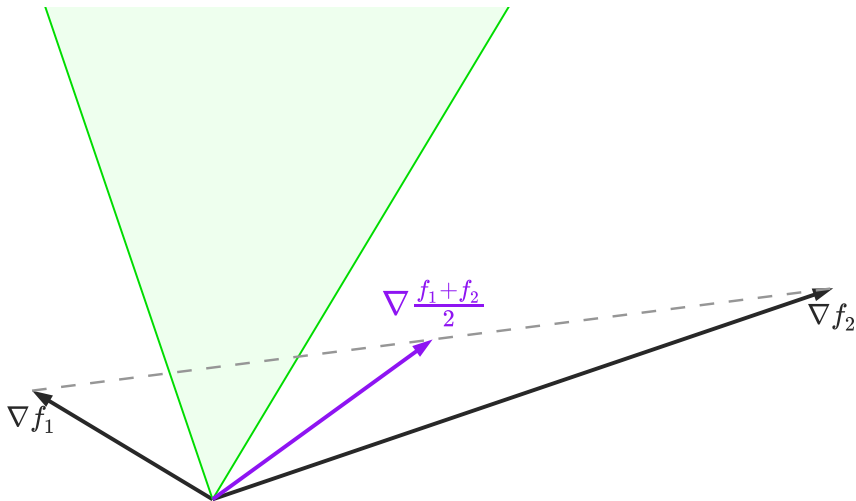
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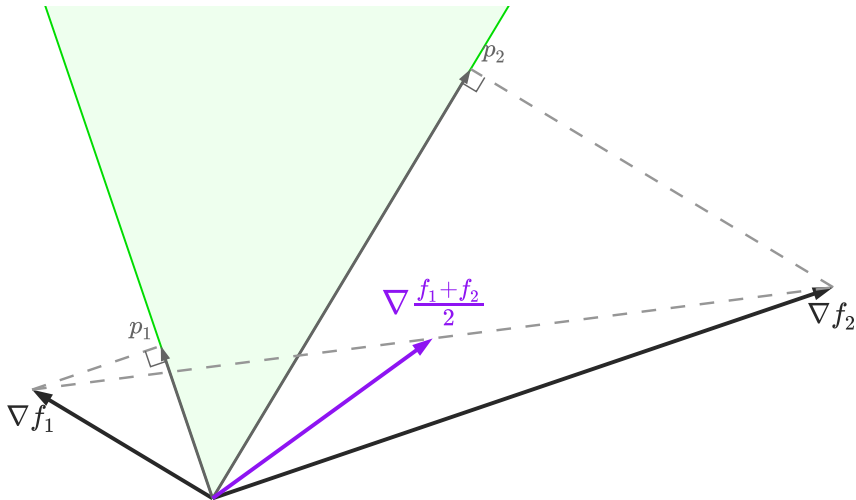
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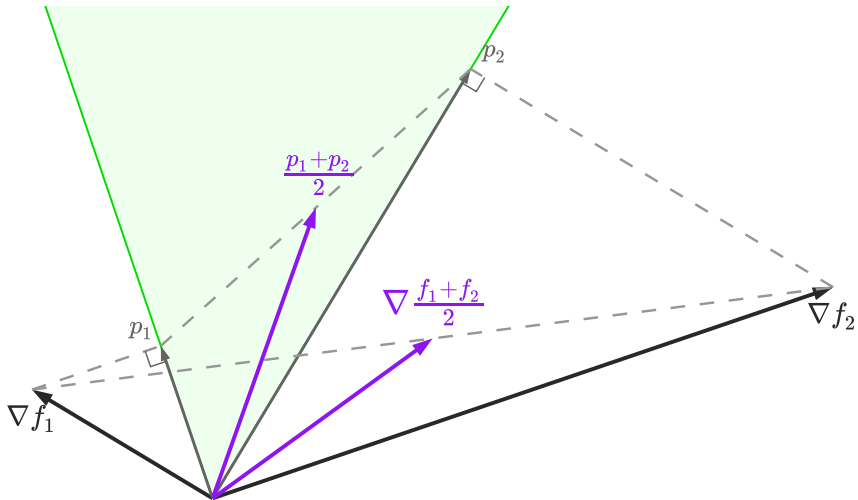
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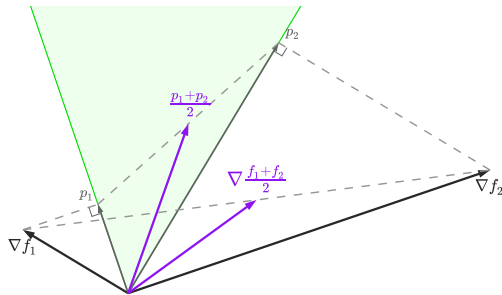


Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\mathbf{y} \in \mathbb{R}^n : \mathbf{0} \preceq J\mathbf{y}\}$.

$$p_i = \arg \min_{\mathbf{y} \in C_J} \|\nabla f_i - \mathbf{y}\|^2$$

$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} p_i$$



TorchJD

TorchJD: core functions

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torchjd	autojac.backward	compute jacobians and accumulate in .jac
torchjd	autojac.jac_to_grad	aggregate .jac fields into .grad

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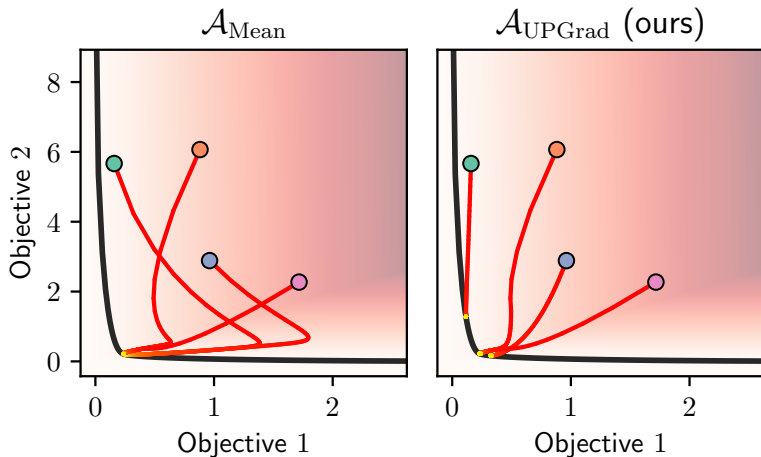
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    optimizer.step()
    optimizer.zero_grad()
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Discussion